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APPLIED ENVIRONMENTAL STATISTICS

SAMPLING STRATEGIES
FOR NATURAL RESOURCES
AND THE ENVIRONMENT

TIMOTHY G. GREGOIRE
HARRY T. VALENTINE

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Chapman & Hall/CRC

Taylor & Francis Group

Boca Raton London New York

Chapman & Hall/CRC is an imprint of the
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CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

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Version Date: 20140313

International Standard Book Number-13: 978-0-203-49888-0 (eBook - PDF)

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*For George M. Furnival,
our mentor, colleague, and friend.*

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Preface

This book is aimed at students and researchers of ecology, natural resources, forestry, or environmental sciences who need to know how to sample natural populations and continuums in a credible manner. The presumed purpose is to estimate, with quantifiable precision, aggregate characteristics of such resources that cannot be measured completely. We present and discuss methods to estimate aggregate characteristics on per unit area basis, say per hectare, as well as on an elemental basis. While some of the sampling designs, e.g., simple random sampling and list sampling, that we present in this book may be found in virtually any text on sampling, others, e.g., randomized branch sampling and 3P sampling, are rather specialized and have evolved specifically for sampling vegetation.

A distinguishing feature of this book is our emphasis on areal sampling designs, including plot sampling, Bitterlich sampling, and line intersect sampling. We also present several recently conceived lesser known areal designs, and we provide solutions to the problem of edge effect, which accompanies the application of many, if not most, areal designs. Our approach to explanation of areal sampling is decidedly geometric, relying heavily upon diagrams and the concept of the inclusion zone.

A second distinguishing feature of this book is our inclusion of sampling designs for continuums. Here we make use of methods of Monte Carlo integration, including crude Monte Carlo, importance sampling, and control variates. These methods originally were used in computer simulation, yet they adapt nicely for the purpose of estimating attributes of continuums such as tree stems, landscapes, lakes, and spans of time.

Because this book is primarily for students of natural resources and the environment, we have chosen to emphasize design and estimation, rather than sampling theory. While we cannot avoid the use of mathematical formulae and expressions, we strive first to provide a conceptual understanding of each sampling design or estimation procedure to which the mathematics apply. For those that are more technically adventurous, we have provided a modicum of derivations and proofs in appendices of chapters. Skipping the technical section will not disadvantage those who lack interest in these seemingly arcane details, but we trust that their mastery will benefit the adventurous by providing a deeper understanding of the underpinnings of sampling theory, estimation, and inference.

As in any sampling text, we impose precise meanings to many common words, and we use some specialized terms that either can be distracting or enlightening, depending on one's disposition. Realizing that the use of this sampling vernacular is an unavoidable necessity in order to establish a common and succinct dialogue, we have attempted to provide a gentle familiarity with essential terms. For example,

Chapter 1 includes a short discussion of what we mean by a ‘population’ and who gets to decide what the population comprises in a particular setting. Words and phrases we feel to be essential to discourse are indicated by *slant type, like this*. Terms to remember also appear together in a box at the end of each chapter.

Short worked examples appear throughout the book, in the hope that they will provide clarity in cases where the symbolic formulas appear mysterious. In virtually all cases, a simple spreadsheet computer program is sufficient to work through each example on one’s own. Graphical displays of data appear throughout the text and we encourage their use as a fundamental step in any analysis of data.

Rarely is there a sampling strategy that is singularly better than any other for a given situation, especially when implementing surveys designed to gather information that will be used to estimate multiple characteristics of the population of interest. Indeed Godambe (1955) established that there is no estimator of the population mean that is best, in a well defined sense. The proof of this result is quite beyond the scope of this book. As we progress through our discussion of sampling and estimation we strive to make comparisons to previously presented methods.

The reader of this text should have some prior knowledge of statistics. We expect familiarity with the normal (Gaussian) and the t distributions, the central limit theorem, elementary notions of probability, the mean and variance of a random variable, and confidence intervals. An introductory course in calculus, or at least a vague memory of one, should provide sufficient foundation for understanding the sections of the text that deal with continuums and Monte Carlo integration.

Material Covered in this Book

Chapter 1 introduces fundamental concepts, and by so doing also establishes some of the vernacular and tenets of probability sampling and design-based inference. In Chapter 2 we develop the notion of a sampling distribution, estimation, and properties of estimators based on the randomization distribution of possible estimates.

In Chapter 3, we introduce five equal probability sampling designs applicable to discretely distributed populations, coupled with various estimators of the population mean value or total value of some attribute of the population. Included in this mix is a discussion of systematic sampling. These topics are then extended to unequal probability sampling designs. Examples from empirical populations are presented to illustrate the applications of these strategies. Chapter 4 parallels Chapter 3 in its presentation of sampling designs and appropriate estimators for continuums.

Stratified sampling is presented in detail in Chapter 5. Related topics which deal with allocation of sampling effort, double sampling to estimate the strata weights, and poststratification are included in this chapter. Generalized ratio and regression estimation is the topic of Chapter 6, which deals more generally with the use of auxiliary information to improve the precision of estimation.

From Chapter 7 onwards we present areal designs that have developed or have been applied in the fields of ecology and natural resource management. Sampling with fixed size plots and quadrats is the subject of Chapter 7. Bitterlich sampling—a.k.a., horizontal point sampling or variable radius plot sampling—is explored in

Chapter 8. Well known and widely used in forestry, there has been considerably less exposure and understanding of the mechanics and utility of the Bitterlich design in other fields. A fixed-population approach to line intersect sampling is presented in Chapter 9. We consider estimators that condition on the orientation of the line transects, as well as unconditional estimators. In all three chapters, special attention is given to edge effects and the possible bias that may ensue.

Chapter 10 draws on the authors' work of the past two decades on the application of Monte Carlo integration to the sampling of continuums. In this chapter we explain how plot sampling, Bitterlich sampling, line intersect sampling, and other areal designs can be formulated as applications of importance sampling. We also deal with edge effect in this context.

Chapter 11 follows with brief presentations of point relascope sampling, variants of horizontal line sampling, transect relascope sampling, ranked set sampling, adaptive cluster sampling, and 3P sampling. Because comprehensive treatments of ranked set and adaptive cluster sampling are available elsewhere, our treatment of these topics is adumbrative rather than comprehensive.

Two-stage sampling designs and cluster designs are considered in Chapter 12. Our treatment of this topic is strategic rather than comprehensive. Two-phase sampling, or double sampling, is also discussed in the book, but rather than devote a separate chapter to this topic, we have chosen to weave it into the fabric of other chapters, where appropriate. Randomized branch sampling, a multistage design, is discussed in Chapter 13. This design conceivably is applicable to any entity that assumes a branched structure, though the design originally was formulated to estimate attributes of individual plants and trees.

Chapter 14, our final chapter, introduces sampling with partial replacement, which is a very economical design for sampling resources on two or more occasions.

Data

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Acknowledgments

"I'm getting tired of citing this book as in preparation!" Tom Lynch, 2006 NEMO meeting

We have fabricated a list of excuses, some of them valid, for missing our initial and several subsequent target dates for publication, but we decided not to print the list, because the book is already long enough. We do, however, thank our editor, Bob Stern, and the staff at Taylor & Francis Group for their forbearance and good humor in the matter of the moving target, and for their assistance in all other matters.

During the process of proposing and writing of this book, we have benefited from reviews, technical comments, material, and support from Dave Affleck, Lucio Barabesi, Ken Brewer, Tom Burk, Ian Cameron, Ann Camp, Mark Ducey, Marian Eriksson, Zander Evans, Steve Fairweather, Lorenzo Fattorini, Jeff Gove, Ed Green,

Linda Heath, Kim Iles, Adrian Lanz, Tom Lynch, Ross Nelson, Anna Ringvall, Andrew Robinson, Frank Roesch, Christian Salas, Oliver Schabenberger, Chip Scott, Goran Ståhl and his students at the Swedish University of Agricultural Sciences, Al Stage, Jim Thrower, Paul Van Deusen, and Mike Williams.

Data for examples and exercises were graciously provided by Jim Barrett, Paul Doruska, Mark Ducey, Teresa Fonesca, Jeff Gove, Dave Hollinger, Mike Lavigne, Doug Maguire, The Natural Resources Management Corporation, Rich Oderwald, Bernard Parresol, Rick Schroeder, Steve Stehman, Harry Wiant, and Daniel Zarin.

Heather Rose, and Joan and Ben Valentine provided several editorial improvements—some would say translations! Linda Heath proofread the entire book and also provided crucial editorial improvements. Kristin Valentine designed our favorite part of the book, the front cover.

We appreciate the supporting network of colleagues, friends, and family who helped us in countless other ways during the preparation of this book.

TGG
HTV

CHAPTER 1

Introduction

1.1 The need for sampling strategies

A sampling strategy combines procedures for selecting a sample from some larger population with procedures for estimating one or more attributes of that entire population from measurements taken on the sample. As its title implies, the focus of this book is on sampling strategies applicable to environmental and natural populations, including continuums. It is this pointedly biological, ecological, and environmental orientation that distinguishes this book from other sampling texts. Without exception, however, the principles of sampling and estimation described here apply equally well to any disciplinary application.

In almost all cases, sampling is conducted because it is impractical, perhaps even impossible, to completely census the entire population of interest without exhausting available resources. Consider, for example, the impossibility of an attempted census of the below-ground biomass of vegetation in the Amazon basin of South America. In this case not only would the enormity of the task deplete national treasuries, but some of the vegetation itself likely would be destroyed in the process.

Moreover, a population census may be fraught with difficulties such as measurement error, undercounting, and erroneous tabulation. Indeed the time it takes to conduct a 100% inventory of a population may be so long that the population itself may change during the process. In this regard, imagine the implausibility of a complete inventory of the number of rhododendron blossoms during spring in the southeastern United States.

A well designed and executed sampling strategy provides a more efficient alternative to a census, where efficiency relates the amount of information obtained to the resources expended. In view of the potential inaccuracies of a population census, a more accurate accounting of the population may in fact be achievable through sampling.

National political campaigns for public office are always attended by pre-election polls of the electorate, with the results of these polls reported publicly. These polls and their results provide commonplace examples of sampling strategies and their outcomes with which everyone is familiar. Less visible than political polls are the numerous demographic, biological, and official surveys conducted on an ongoing basis by governmental, scientific, and economic bureaus, to name a few sources. Like an electoral poll, a sample, for whatever purpose, is a survey of some portion of a larger group from which one tries to infer meaningful quantitative information descriptive of the larger group, which we designate as the *population* of interest.

We have already insinuated that a *population* is the entire collection of subjects

or objects about which we wish to know something. The sampler must decide which collection of objects or subjects constitutes the population of interest. In one case or study it might be a particular species of plant or all plant species in a geographic region; in another case, it might be plants of a particular species and age combination; and in yet another case, attention might be limited to just the vigorous plants, where vigor is defined in some unambiguous and identifiable way. For notational convenience we shall use the symbol $\mathcal{P}$ throughout this book to refer to the population of interest.

Need the population be discrete, or can it be continuous? The seed cones on a Norway spruce (*Picea abies* L.) tree is an example of a discrete population, whereas the forest floor, upon which the cones eventually fall, is an example of a continuous population or continuum. The members of discrete populations are customarily called *elements* or *units*. A continuous population, by contrast, comprises infinitely many *points*, for example, the points comprising the surface of a forest floor or the surface of a lake. More sampling theory and practice has been directed towards discrete populations than continuous populations. Sampling theory for the latter nevertheless has advanced in the disciplines of Monte Carlo integration (Rubinstein 1981), plane sampling (Quenouille 1949; Dalenius et al. 1961), remote sensing (Koop 1990), and geology (de Gruijter & ter Braak 1990). With recent emphasis on environmental monitoring, sampling strategies for continuous populations has garnered increased attention (Cordy 1993; Overton & Stehman 1993; Stevens 1997). Within forestry, Gregoire et al. (1986) defined a continuous population that comprised the continuum of cross sections along the central axis of the bole of a tree.

The population of interest nearly always is determined by the research or informational objectives of the survey, and ideally this is the population that is targeted for sampling. The important point is that the estimate derived from a sample of the targeted population pertains only to that population. To the extent that the intended population differs from that targeted by the sampling design, extrapolation of the sample results to the intended population is inherently risky and may be subject to legitimate criticism. An example may help to clarify the distinction between the intended and targeted population. In the course of sampling to estimate foliar area per hectare of all Amazonian vegetation, suppose we sample plants in such a way that only vegetation with woody stems can be selected and measured. In this scenario, only the woody vegetation is targeted for sampling, and the herbaceous component of the population of interest is given zero chance of being included in the sample. We cannot extend this estimate to include foliar area of herbaceous vegetation without a leap of faith or without a demonstration that the resulting bias is acceptably small. In this text we leave faith to theologians and concern ourselves only with sampling strategies that are statistically defensible, objective, and if not design-unbiased, then estimation with bias that is negligible as the size of the sample increases.

1.2 A medley of sampling scenarios

In the previous section we mentioned the sampling of vegetation in the Amazon for the purpose of estimating foliar area. This section presents a suite of scenarios to illustrate the broad spectrum of uses of probability sampling applied to ecological and natural resources.

1.2.1 Sampling for biomass estimation

Biomass of living vegetation in a region is important for a host of reasons relevant to ecological inquiry or resource management. It might, for example, serve to indicate the region's nutritive capacity and its capability to support vegetation, or it may reflect a site's response following management action such as fertilization, burning, irrigation, or harvesting. Measurement of aboveground biomass of a plant is usually a chore, often involving drying of the plant material to reduce its moisture content to an acceptably small level. For very large plants, say mature trees, the collection, drying, and measurement of biomass of a single individual is an arduous task owing to the sheer amount of living tissue aboveground. While the amount of living tissue below ground may be less, the task of collecting this material for purpose of measurement is challenging because of its difficult access. Moreover, the actual measurement of biomass kills the plant. A well planned sampling of aboveground biomass decreases the amount of effort required, without extirpating the population of interest.

If the population of interest comprises lesser vegetation—shrubs and herbaceous plants—then the size of each individual presents far less of a challenge to measurement of biomass, but this is supplanted by a likely increase in the number of individuals, perhaps a manyfold increase. Here again it makes sense to consider measuring but a portion of the population, and using that sample to estimate the biomass of the entire population. There are innumerable ways in which such a sample can be selected. Regrettably, there is no singular way to collect a sample that is optimal for all populations, circumstances, and information needs. This book presents a few alternative strategies—with comparative advantages and disadvantages which can vary greatly from one setting to another.

1.2.2 Sampling to estimate composition

Providing that you can recognize and distinguish among different plant species in your backyard, it likely would be a straightforward task to tabulate the number of plants of each species and then compute the proportional representation of each species on your property. Doing similarly for the White Mountain National Forest in northern New Hampshire would require either a huge crew of trained labor, or else a reliance on some method of sampling coupled with some way to transform the observations acquired in the sample into estimates applicable to the population. Indeed, even in your backyard, if grasses and forbs were included in your definition of the population, a sampling strategy likely would be required there, also.

In this book, discussions of sampling design involves issues such as the manner in which elements or subgroups are selected for the sample, as well as how large

the sample should be. Discussions of estimation involve alternative uses of the quantitative and qualitative information in a sample to arrive at an estimate of an attribute of the population of interest, e.g., the proportion of the population which is coniferous. Discussions of inference embrace issues relating to the reliability of estimation.

1.2.3 Sampling for cover information

How much of the desert floor is covered by pieces of petrified wood in the Petrified Wood National Park in Arizona? Officials of the National Park Service were interested in this question because the amount of petrified wood in its natural state seemed to be diminishing in high traffic areas of the park: some visitors, in defiance of posted prohibitions of the practice, pilfer pieces of petrified wood to take home as souvenirs. In this situation, it seemed to make sense to adopt one sampling strategy for the subpopulation of petrified wood wherein pieces were the size of one's fingernail or smaller, a different sampling method for that subpopulation consisting of log-size pieces, and yet another sampling method for intermediate-size pieces. This technique of stratifying the population into different subpopulations, and then tailoring the sampling design within each subpopulation is commonly employed and can produce tremendous efficiencies of effort. We treat stratified sampling at some length in this book.

Estimation of plant canopy cover has long been of interest to ecologists (cf., Canfield 1941). Curiously, the line intercept technique of sampling canopy cover can be applied, as well, to the estimation of area of forest gaps. We provide details in Chapter 9.

1.2.4 How many trees in Sweden?

Dalenius (1957) mentions an early effort to obtain a probability sampling of the timber resource in Sweden in 1910–1912: it involved counting all trees within strips of land located systematically along parallel lines extending from Sweden's western border to its eastern border. The Varmland survey, as it was called, has historical significance for a number of reasons: from a statistical viewpoint, "... [t]his appears to be the first Swedish survey where the problem of measuring the degree of representativity was approached in terms of probability calculus.... The pioneer character of the Varmland survey was apparent to the forestry statisticians of the time [and it] played a decisive role for the development of sample survey practice in forestry statistics and related fields" (Dalenius 1957, p. 45, 49). Indeed, it served as a pilot survey for subsequent national forest inventories in Sweden initiated in 1923*.

By the start of the 21st century, technology has reduced the requisite field labor and time of large regional and national forest inventories, although the scope of such inventories has expanded enormously. Typically, remotely sensed data from satellite images or aerial photography will be used to stratify the region into noncontiguous

* In 2000, the Swedish National Forest Inventory estimated that Sweden contains 65,735,000,000 trees with heights greater than 1.3 m

areas of homogeneous cover, such as water, forest land, non-forest land. The intensity of sampling within each stratum will vary according to stipulated, or perhaps implicitly stated, needs of accuracy, and a subsample of ground locations will be selected for visitation by a field crew. At each ground location, a few measurements may be taken, such as a count of the number of trees in the vicinity of the ground location, the depth of the A and B soil horizons, or the degree of slope. Or, in consideration of the expense of placing a crew in the field, there may be a few dozen, possibly hundreds, of measurements of vegetative, wildlife, topographic, hydrologic, and geologic condition. Moreover, each ground location may serve as a point anchor for a lattice of additional locations around that point. Within the lattice, the measurement protocol may vary for some features, and may be constant for others. We mention this scenario to highlight the fact that many sample surveys of a population embrace multiple attributes, have multiple layers and phases which serve to refine the sample selection process, the estimation process, or both (Nusser & Goebel 1997; Nusser et al. 1998; Rennolls 1989; Rudis 1991; Smith & Aird 1975; FAO 1990; FHM 1998). This is true, also, of many large demographic surveys.

1.3 Probability sample

Our initial focus, in this book, is on sampling designs that select probability samples from discrete populations. A probability sample is underpinned by two probabilities: the selection probability of the sample and the inclusion probabilities of the units in the sample. We discuss the former first.

1.3.1 Selection probability of a sample

Each and every probability sample has a selection probability that is deducible from the strictures of the sampling design or protocol. We adopt the shorthand notation $p(s)$ to denote this selection probability of a sample, s . Although both simple random

ASIDE: When sampling for the purpose of estimating one or more quantitative attributes of a population, one fundamentally aims to select a sample that is representative of the population. What makes a sample representative? The answer to this deceptively simple question depends, of course, on what one means by representative. As Kruskal & Mosteller (1979a,b,c) explore in a very readable and informative series of articles, the term ‘representative sampling’ admits to a wide range of use not only across the scientific and non-scientific literature, but also within the considerably narrower statistical literature. Indeed in the latter, Kruskal & Mosteller (1979c) elaborate on nine distinctive meanings that have appeared for the term representative sampling. We do not intend to add yet another usage, but will instead follow the sage advice of W.E. Deming, as quoted by (Jones 1958), by phrasing our discourse in terms of probability sampling.

Table 1.1 *Distinct samples of size $n = 2$ from a population with $N = 6$ elements.*

Sample	Elements	Sample	Elements	Sample	Elements
s_1	$\mathcal{U}_1, \mathcal{U}_2$	s_6	$\mathcal{U}_2, \mathcal{U}_3$	s_{11}	$\mathcal{U}_3, \mathcal{U}_5$
s_2	$\mathcal{U}_1, \mathcal{U}_3$	s_7	$\mathcal{U}_2, \mathcal{U}_4$	s_{12}	$\mathcal{U}_3, \mathcal{U}_6$
s_3	$\mathcal{U}_1, \mathcal{U}_4$	s_8	$\mathcal{U}_2, \mathcal{U}_5$	s_{13}	$\mathcal{U}_4, \mathcal{U}_5$
s_4	$\mathcal{U}_1, \mathcal{U}_5$	s_9	$\mathcal{U}_2, \mathcal{U}_6$	s_{14}	$\mathcal{U}_4, \mathcal{U}_6$
s_5	$\mathcal{U}_1, \mathcal{U}_6$	s_{10}	$\mathcal{U}_3, \mathcal{U}_4$	s_{15}	$\mathcal{U}_5, \mathcal{U}_6$

sampling and systematic sampling designs (discussed in Chapter 3) ensure that $p(s)$ is identical for each of the possible samples, there is no restriction that the design of a probability sample must ensure that all possible samples be equally likely.

Under most sampling designs, the number of possible samples will far outnumber the elements in the population, because many sampling designs permit a partial overlap in composition, i.e., two distinct samples may have some population elements in common. In practice, it never is necessary to enumerate all possible samples and the composition of each, but we do so here to illustrate this last point. In a well-known primer, *Basic Ideas of Scientific Sampling*, Stuart (1962) examined a six element population with members $\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3, \mathcal{U}_4, \mathcal{U}_5, \mathcal{U}_6$. Consider now the various samples each with two distinct elements that may be drawn from this population; these are shown in Table 1.1. Using the symbol Ω to denote the number of possible samples, it is evident from the tabulated display that for this example $\Omega = 15$. While we have shown the first two samples, s_1 and s_2 , as comprising the elements $\{\mathcal{U}_1, \mathcal{U}_2\}$ and $\{\mathcal{U}_1, \mathcal{U}_3\}$, respectively, the ordering of the samples in this enumeration is arbitrary. Indeed, the sample labels, s_1, s_2, s_3 , and so on, are introduced merely as a notational convenience to help us distinguish one possible sample from another in our discussion. Another person may choose to enumerate the list of possible samples in a different order, one that would lead, for example, to the designation of sample s_3 as containing elements $\{\mathcal{U}_3, \mathcal{U}_4\}$.

In this example from Stuart, we have not stipulated the probability of obtaining, say, sample s_7 , i.e., we have not asserted anything about the value of $p(s_7)$ except that it is deducible for any particular probability sampling design. Nonetheless, it is possible to reason that $p(s_7) > 0$, because otherwise $p(s_7)$ must be identically zero, a result that is impossible if, indeed, s_7 is one of the possible samples. Likewise, $p(s_7) < 1$, because otherwise $p(s_7) = 1$, a result which implies that no other samples are possible. We elaborate on this point in order to establish that, under a probability sampling design, $0 < p(s) < 1$ not only for s_7 but for each possible sample. Furthermore, remembering that Ω represents the number of possible samples obtainable under a given design, then $1 = \sum_{k=1}^{\Omega} p(s_k)$.

For discrete populations the magnitude of Ω will depend both on the number of population elements selected for the sample, n , and the number of elements in the population, N . An exception to this generality happens when the sampling design is one that permits the size of the sample, n , to be random; i.e., n cannot be determined

in advance of sampling, in which case Ω will depend on N but not n . For this reason, among others, we will be careful to distinguish whether the sample size, n , for each sampling design presented in this book is random or not; the latter will be mentioned as a fixed- n design, whereas the former will be called a random- n design. For infinitely large populations ($N = \infty$) and for continuously distributed populations, Ω likewise will be infinite.

If a probability sample of size n is to be chosen from a population of size N , then the protocol of the design, or more simply stated, the design itself, will determine the probability of obtaining each of the Ω possible samples. There are a few common sampling designs that ensure that $p(s)$ is identical for each of the Ω possible samples, in which case it is evident that $p(s) = 1/\Omega$ for all s permissible under the design. There are many more sampling designs which cause $p(s)$ to vary among samples composed of different elements. Moreover, $p(s)$ for any particular sample, say s_4 from Table 1.1, will depend on the design: the value of $p(s_4)$ under one sampling design will differ from its value under another sampling design, in general.

1.3.2 Inclusion probability

Each element of the population must have a nonzero probability of being included in a sample, which implies that each element must appear in at least one of the possible samples permissible under the design. Otherwise, the sampling is not probability sampling.

We distinguish between the sample probability, $p(s)$, of selecting a particular sample—when sampling according to a specific sampling design—from the probability of including a particular population element into a sample, s , under that design. This distinction perhaps can be made clearer by means of a short example. For the $N = 6$ population whose $\Omega = 15$ possible samples, each of $n = 2$ distinct elements, that are enumerated in Table 1.1, element $\mathcal{U}_5$ appears in samples s_4 , s_8 , s_{11} , s_{13} , and s_{15} . Without being specific for the moment as to the sampling design, suppose that the corresponding sample probabilities are denoted as $p(s_4)$, $p(s_8)$, $p(s_{11})$, $p(s_{13})$, and $p(s_{15})$. Let π_5 represent the overall probability of including $\mathcal{U}_5$ in a randomly chosen sample under this design. Upon reflection it can be deduced that $\pi_5 = p(s_4) + p(s_8) + p(s_{11}) + p(s_{13}) + p(s_{15})$. In the current parlance, this overall probability of including an element of the population in a randomly selected sample is its *inclusion probability*. Because the sample probabilities, $p(s)$, under one design will differ from those determined for another design, so too will the inclusion probability of $\mathcal{U}_5$. Using a more general notation, if the inclusion probability of the k th population element, $\mathcal{U}_k$, is denoted as π_k , then it must be true that

$$\pi_k = \sum_{s \ni \mathcal{U}_k} p(s), \quad (1.1)$$

where the notation $s \ni \mathcal{U}_k$ indicates that the summation extends over all samples of which $\mathcal{U}_k$ is a member. We emphasize, again, that because π_k is determined by the sampling design, its value under one design will not be identical, in general, to its value under another design.

ASIDE: Some sampling designs permit the same population element to be selected two or more times for the sample. In essence, once the first sample element is selected from the population, that element is replaced in the population, so that it has the same probability of being selected on the second, and succeeding, selections. Evidently such ‘with-replacement’ sampling designs imply a sequential selection process, and it is possible to discern yet a different probability than either of the two mentioned above. The *selection probability* of a population element, say $\mathcal{U}_k$, is the probability that this unit will be selected in each of the sequenced selections under a with-replacement sampling design. We will discuss these selection probabilities of population elements in greater detail when we first present a with-replacement sampling design in Chapter 3. We also will show the relationship between a unit’s selection probability and its overall inclusion probability at that point in the text.

Another defining characteristic of a probability sample is that the inclusion probability of any unit of the population be deducible. When sampling from a continuously distributed population, the corresponding notion is that the probability density be deducible (we discuss probability densities at length in Chapter 4). Some probability sampling designs ensure that the inclusion probability is constant for all elements of the population, but most do not. For many designs presented in this book, the inclusion probability of a population element may be computed without having to enumerate all possible samples and computing the $p(s)$ of each.

A particular probability sample may not be a miniaturized version of the population, nor will it necessarily comprise elements that are, in some vague sense ‘typical’ elements of the population. While there are devices, such as stratification and ordering, that can be employed to more nearly match the sample composition to that of the population, the merit of probability sampling lies chiefly with its guarantee of desirable behavior on average, not with desirable behavior of each and every sample that can be selected.

1.3.3 Sampling frame

A sample is selected with the aid of a *sampling frame*, which we regard as any mechanism that allows one to identify and select elements of the population. It might be as simple as a list, somehow numbered or labeled, of all population elements or groups of elements. The frame might be based on area, such that all elements within a selected area are observed; it might constitute a labeled list of elements along with one or more values of auxiliary characteristics corresponding to each. The units constituting the sampling frame may be identical to the population units, or it might contain groupings of units. The collection of population elements included within the sampling frame effectively constitutes the target population, and thus it is a matter of some consequence that the target population coincides with the population about which one wishes to draw an inference.

1.4 Inference

Objective selection is a hallmark of probability sampling. Estimates that derive from probability samples are free from the distorting influences that can be insinuated by subjective selection. Nonetheless, it is natural to question how reliable an estimate from a probability sample can be. How different might another estimate provided from a second sample be from the estimate provided by the first sample? Would one's estimate coincide with the population value in the hypothetical situation where the size of the sample matches the size of the population? Providing answers to questions like these leads one into the area of *statistical inference*.

In common with other established texts on sampling methods, such as the superb works by Cochran (1977) and Särndal et al. (1992), we stay entirely within the realm of *design-based inference*. Succinctly stated, properties of an estimator are deduced from the distribution of all estimates possible under the stipulated sampling design. In the design-based framework the population of interest is regarded as a fixed, not a random, quantity. If, for example, the attribute of interest is aboveground biomass of the N trees in a forest, then the fixed-population concept implies that the biomass of each tree in the population of N trees is likewise a fixed quantity, perhaps unknown until that tree is selected into the sample and measured. For purposes of statistical inference in the design-based framework, nothing is assumed about the manner in which the population elements, the trees, are distributed nor how the characteristics are distributed. In other words, there is nothing random about the population being sampled. Randomness enters only through the sample selection procedure, i.e., *sample design*.

To illustrate the notion of a fixed population, consider the population consisting of all the Douglas-fir trees in British Columbia taller than 1 m. Our aim is to estimate the foliage biomass of this population. Douglas-fir trees can be found along the coast and in the mountains; they grow and compete for nutrients and water among other Douglas-fir trees and among trees of other species. One can imagine grouping together all Douglas-fir trees that are 10 years old, 11 years old, and so on, and then examining how the numbers of trees are distributed across the spectrum of age classes. All of these factors, and numerous others that are deducible from the biology of Douglas-fir growth and that may be related to the amount of foliage supported on each tree, are essentially irrelevant in the system of design-based inference we expound in this text. That is not to say that auxiliary information of this sort cannot be used to great advantage in the design of the sampling procedure, or weaved into the estimator of total foliage, or both. The point of emphasis here is that the process by which we assess the accuracy and reliability of an estimator in no way depends on the aptness of any of these presumed relationships.

By contrast, model-based inference rests on an assumed structure for the population, i.e., a population model. Cassel et al. (1977) and Chaudhuri & Stenger (1992) provide extensive, and quite mathematically rigorous, coverage of model-based inference. A useful way to distinguish these two modes of inference is to remember that in the design-based approach the sample is a realization of a random process, whereas in the model-based approach the population is regarded as a realization of

a random process. Contrasts of the two modes of inference have been penned by Särndal (1978), de Gruijter & ter Braak (1990), and Gregoire (1998).

Adherents of the model-based mode of inference cite its congruence with much else that is practiced in statistical analysis: a model is postulated and, conditionally on the veracity of that model, one draws conclusions about a population, a relationship, or a stochastic process. Adherents of the design-based mode of inference cite its freedom from a priori assumptions, i.e., inference is valid irrespective of the concordance of the presumed model with reality. Both inferential approaches have merit, and it is important that the basis or mode of inference at all times be unequivocally stated. Therefore, when one characterizes properties of an estimator of a population attribute it is important to be quite clear whether these properties are assessed with regard to the reference distribution of all possible estimates permitted by the sampling design, or by the stochastic relationship presumed for the population. An estimator that is unbiased under one mode of inference may well be biased under the other.

1.5 Population descriptive parameters

1.5.1 Discrete populations

Let y_k denote the value of an attribute of interest, which is associated with the k th unit, $\mathcal{U}_k$, of a discrete population. The population total obtains by summing the attribute across all N units, i.e.,

$$\tau_y = \sum_{k=1}^N y_k.$$

The population total, τ_y , is a *population descriptive parameter* and often a parameter of interest in resource management or scientific inquiry. For example, in the context of the previous section, y_k might be the foliar biomass of the k th tree in the sampling frame, in which case, τ_y is the total biomass contained in the population of N trees.

The population total often is the *target parameter*, i.e., the population descriptive parameter to be estimated. An alternative target parameter is the population mean, μ_y , i.e.,

$$\mu_y = \frac{\tau_y}{N} = \frac{1}{N} \sum_{k=1}^N y_k$$

Generally, μ_y is the average amount of attribute per unit. In the biomass context, μ_y obviously is the average biomass per tree.

There may be occasions where a sample is selected for the purpose of estimating not only τ_y or μ_y , but also for estimating population parameters of other attributes. For example, one might be interested in estimating the average biomass per tree, the average tree height, and the average tree diameter. As needed, we will use x , z , and other letter symbols to denote additional attributes of interest. In similar fashion to y_k , x_k will denote the value of the x attribute on the k th unit, and so on.

Another population descriptive parameter of interest sometimes is the ratio of two

ASIDE: Many will find the notation we use esoteric at first, yet it is a notation that is necessary for precise discourse. The quantitative formula of a population attribute, such as $\tau_y = \sum_{k=1}^N y_k$, unambiguously defines the attribute and makes possible a symbolic expression with which the value of this population descriptive parameter may be ascertained, in principle. In practice, we rely on sampling to estimate its value. A population descriptive parameter is one that can be expressed quantitatively in some fashion: for example, the proportion of foliage in a well-defined color class is estimable, whereas the overall color of foliage is not estimable by the sampling methods we expound.

population parameters, e.g.,

$$R_{y|x} = \frac{\tau_y}{\tau_x} = \frac{\mu_y}{\mu_x}.$$

The difference between the minimum and maximum value of y in a population is the *range*, and it is an obvious measure of the spread of values in the population. The range is yet another population descriptive parameter, however, it is one that rarely is of interest. An alternative quantitative measure of how dispersed the values of y are in the population is the *variance* of y , symbolized by σ_y^2 , and expressed by

$$\sigma_y^2 = \frac{1}{N-1} \sum_{k=1}^N (y_k - \mu_y)^2.$$

If one regards $y_k - \mu_y$ as the ‘distance’ of y_k from the population mean, then a useful interpretation of σ_y^2 is as the average ‘squared distance,’ approximately, between individual observed values and their average value. The variance of y also can be interpreted literally: it is a measure of how much the y_k s vary around their mean. The Chapter 1 Appendix provides two alternative, but algebraically equivalent, expressions for σ_y^2 .

Because σ_y^2 is a function of squared values of y , its unit of measure is also the square of the unit of measure of y . For example, if y is a measure of weight in kg,

ASIDE: Various authors define the population variance differently as

$$\sigma_y^2 = \frac{1}{N} \sum_{k=1}^N (y_k - \mu_y)^2$$

which differs by a factor of $\frac{N-1}{N}$ from the definition given above, e.g., Cochran (1977). Moreover, some authors will use a symbol other than the Greek letter σ when defining the population variance in the same fashion as we have done above, e.g., Särndal et al. (1992, p. 39) symbolize it as S_{yU}^2 .

then σ_y^2 is expressed in kg^2 . In contrast, the *standard deviation* of y , namely σ_y , has the same units of measure as y .

At times it is convenient to express variability relative to some benchmark value. The most commonly used benchmark value is the mean, μ_y , and it is used in the population descriptive parameter known as the *coefficient of variation*, γ_y , which is expressed as

$$\gamma_y = \frac{\sigma_y}{\mu_y}$$

or in percentage terms as

$$\gamma_y = \frac{\sigma_y}{\mu_y} 100\%.$$

The coefficient of variation is a meaningful descriptive parameter only when $y > 0$.

Example 1.1

Consider the following three-unit population:

Unit	y	x
	(kg)	(cm^2)
$\mathcal{U}_1$	2	13
$\mathcal{U}_2$	9	25
$\mathcal{U}_3$	17	18

The following results are easily verified:

$$\begin{array}{ll} \tau_y = 28 \text{ kg} & \tau_x = 56 \text{ cm}^2 \\ \mu_y = 9.3 \text{ kg} & \mu_x = 18.6 \text{ cm}^2 \\ \sigma_y^2 = 56.3 \text{ kg}^2 & \sigma_x^2 = 36.3 \text{ cm}^4 \\ \sigma_y = 7.5 \text{ kg} & \sigma_x = 6.0 \text{ cm}^2 \\ \gamma_y = 80.4 \% & \gamma_x = 32.3 \% \end{array}$$

Verify that the same value for $R_{y|x}$ is obtained regardless of whether it is computed as τ_y/τ_x or as μ_y/μ_x .

1.5.2 Continuous populations

Continuous populations do not naturally divide into discrete units, nor do they lend themselves to simplistic description. For our purposes, a continuous population is a domain of integration, $\mathcal{D}$, comprising infinitely many points. In Chapter 4, we discuss sampling in a one-dimensional domain comprising an interval on a line; a two-dimensional domain comprising a bounded area on a plane; and a three-dimensional domain comprising the volume of a container. Each point in a domain can serve as a ‘sample point.’ An attribute of interest, which may extend continuously

along, across, or throughout the domain, depending on the number of dimensions, has density $\rho(\mathbf{x})$ at point $\mathbf{x}$. In one dimension, the attribute density typically is an amount of attribute per unit length or per unit time. In two and three dimensions, an attribute density typically is an amount attribute per unit area and per unit volume, respectively. Consider, for example, the volume of litter on a forest floor. The attribute density is litter volume per unit land area ($\text{m}^3 \text{m}^{-2}$), which reduces to litter depth (m) at a location point, $\mathbf{x}$.

The total amount of attribute extant over the domain of integration is a population descriptive parameter. By definition, this total is equivalent to integral of the attribute density

$$\tau_\rho = \int_{\mathcal{D}} \rho(\mathbf{x}) \, d\mathbf{x}.$$

The subscript ρ indicates that the total, τ_ρ , obtains from the integration of an attribute density over a continuous domain. It also serves to distinguish the continuous total from a discrete total, τ_y , which obtains from a summation of N attribute values.

Let D be the size—the length, area, or volume—of domain $\mathcal{D}$, i.e.,

$$D = \int_{\mathcal{D}} d\mathbf{x}.$$

The mean attribute density across the domain is

$$\mu_\rho = \frac{\tau_\rho}{D}$$

The variance of the attribute density across the domain is

$$\sigma_\rho^2 = \frac{1}{D} \int_{\mathcal{D}} [\rho(\mathbf{x}) - \mu_\rho]^2 \, d\mathbf{x}.$$

Ratios and coefficients of variation are defined as in discrete populations.

1.6 Historical note

In the opening decades of the twentieth century there was considerable debate as to just how sampling should be conducted. Much of this debate compared the merits of a purposive selection procedure to a probabilistic one, and indeed served to highlight the ambiguity surrounding the term representativeness. While there has been recent renewed interest in purposive selection, unquestionably the latter half of the century has witnessed the near dominance of probability-based sampling in scientific and public forums. In large part this is due to an exceedingly influential paper delivered to the Royal Statistical Society by Neyman (1934). In the words of Bellhouse (1988) the major reason why the paper

“...provides a paradigm in the history of sampling is that a theory of point and interval estimation is provided under randomization that breaks out of an old train of thought and opens up new areas of research.”

In addition to the Bellhouse (1988), Hansen et al. (1985) have written a very readable overview of the development of sampling ideas in the twentieth century.

Finally, a dated but still relevant monograph, *The Principles of Sampling*, by Cochran et al. (1954) is worthwhile reading for any who seek to master the techniques presented in this book.

1.7 Terms to remember

Coefficient of variation	Probability sampling
Design-based inference	Sampling design
Elements, units	Sampling frame
Inclusion probability	Sampling strategy
Model-based inference	Standard deviation
Population	Variance

1.8 Appendix

1.8.1 Alternative expressions for σ_y^2

By expanding the squared term, $(y_k - \mu_y)^2$, within the summation of the expression for σ_y^2 , and then collecting terms, one gets

$$\sigma_y^2 = \frac{1}{N-1} \left[\left(\sum_{k=1}^N y_k^2 \right) - N\mu_y^2 \right].$$

Taking this last expression and substituting τ_y/N for μ_y yields

$$\sigma_y^2 = \frac{1}{N-1} \left[\left(\sum_{k=1}^N y_k^2 \right) - \frac{\tau_y^2}{N} \right].$$

Bibliography

- Affleck, D. L. R., Gregoire, T. G., & Valentine, H. T. 2005. Design unbiased estimation in line intersect sampling using segmented transects. *Environmental and Ecological Statistics*, **12**(2), 139–154.
- Affleck, D. L. R., Gregoire, T. G., & Valentine, H. T. 2006. Edge effects in line intersect sampling with segmented transects. *Journal of Agricultural, Biological, and Environmental Statistics*, **10**, 460–477.
- Anonymous. 1999. *Summary Report 1997 National Resources Inventory*. Tech. rept. United States Department of Agriculture, Natural Resources Conservation Service.
- Assmann, E. 1970. *The Principles of Forest Yield Study*. Oxford: Pergamon Press.
- Avery, T. E., & Burkhart, H. E. 2002. *Forest Measurements*. 5 ed. Series in Forest Resources. Boston: McGraw-Hill.
- Bankier, M. 1988. Power allocations: determining sample sizes for subnational areas. *The American Statistician*, **42**, 174–177.
- Barabesi, L. 2003. A Monte Carlo integration approach to Horvitz-Thompson estimation in replicated environmental designs. *Metron*, **LXI**, 355–374.
- Barabesi, L. 2004. Replicated environmental sampling designs and Monte Carlo integration methods: two sides of the same coin. In: *Proceedings of the XLII Meeting of the Italian Statistical Society*. Bari: Italian Statistical Society.
- Barabesi, L., & Fattorini, L. 1998. The use of replicated plot, line and point sampling for estimating species abundance and ecological diversity. *Environmental and Ecological Statistics*, **5**, 353–370.
- Barabesi, L., & Pisani, C. 2004. Steady-state ranked set sampling for replicated environmental sampling designs. *Environmetrics*, **15**, 45–56.
- Barrett, J. P., & Nutt, M. E. 1979. *Survey Sampling in the Environmental Sciences: A Computer Approach*. Wentworth, NH: COMPress, Inc.
- Bartlett, R. F. 1986. Estimating the total of a continuous population. *Journal of Statistical Planning and Inference*, **13**, 51–66.
- Bauer, H. L. 1936. Moisture relations in the chaparral of the Santa Monica Mountains, California. *Ecological Monograph*, **6**(3), 409–454.
- Bebber, D. P., & Thomas, S. C. 2003. Prism sweeps for coarse woody debris. *Canadian Journal of Forest Research*, **33**, 1737–1743.
- Bebbington, A. C. 1975. A simple method of drawing a sample without replacement. *Applied Statistics*, **24**, 136.

- Beers, T. W. 1962. Components of forest growth. *Journal of Forestry*, **60**, 245–248.
- Beers, T. W., & Gingrich, S. F. 1958. Construction of cubic-foot volume tables for red oak in Pennsylvania. *Journal of Forestry*, **56**, 210–214.
- Beers, T. W., & Miller, C. I. 1964. *Point Sampling: Research Results, Theory and Application*. Agricultural Experiment Station Bulletin No. 786. Lafayette, Indiana: Purdue University.
- Bell, J. F., & Alexander, L. B. 1957. *Application of the variable plot method of sampling forest stands*. Tech. rept. 30. Oregon State Board of Forestry.
- Bell, J. F., Iles, K., & Marshall, D. D. 1983. Balancing the ratio of tree count-only sample points and VBAR measurements in variable plot sampling. Pages 699–702 of: Bell, J. F., & Atterbury, T. (eds), *Renewable resource inventories for monitoring changes and trends*. Corvallis, Oregon: College of Forestry, Oregon State University.
- Bellhouse, D. R. 1985. Computing methods for variance estimation in complex surveys. *Journal of Official Statistics*, **1**(3), 323–329.
- Bellhouse, D. R. 1988. A brief history of random sampling methods. Pages 1–14 of: Patil, G. P., & Rao, C. R. (eds), *Handbook of Statistics, Volume 6*. Amsterdam: North Holland.
- Bethel, J. 1989. Sample allocation in multivariate surveys. *Survey Methodology*, **15**, 47–57.
- Bethlehem, J. G., & Keller, W. J. 1987. Linear weighting of sample survey data. *Journal of Official Statistics*, **3**, 141–153.
- Bissell, A. F. 1986. Ordered random selection without replacement. *Applied Statistics*, **35**, 73–75.
- Bitterlich, W. 1949. Die Winkelzählprobe. *Allgemeine Forst- und Holzwirtschaftliche Zeitung*, **59**(1/2), 4–5.
- Bitterlich, W. 1984. *The Relascope Idea: Relative Measurements in Forestry*. Slough, England: Commonwealth Agricultural Bureaux.
- Bliss, C. I. 1941. Statistical problems in estimating populations of Japanese beetle larvae. *Journal of Economic Entomology*, **34**, 221–232.
- Bormann, F. H. 1953. The statistical efficiency of sample plot size and shape in forest ecology. *Ecology*, **34**(3), 474–487.
- Brewer, K. R. W. 2002. *Combined Survey Sampling Inference: Weighing Basu's Elephants*. London: Arnold.
- Brewer, K. R. W., & Gregoire, T. G. 2000. Estimators for use with Poisson sampling and related selection procedures. Pages 279–288 of: *Proceedings of the Second International Conference on Establishment Surveys*. Alexandria, Virginia: American Statistical Association.
- Brewer, K. R. W., & Hanif, M. 1982. *Sampling With Unequal Probabilities*. New York: Springer-Verlag.

- Brooks, J. R. 2006. An evaluation of big basal area factor sampling in Appalachian hardwoods. *Northern Journal of Applied Forestry*, **23**(1), 62–65.
- Brown, J. A. 2003. Designing an efficient adaptive cluster sample. *Environmental and Ecological Statistics*, **10**, 95–105.
- Brown, J. K. 1974. *Handbook for inventorying downed woody material*. Tech. rept. INT-16. USDA Forest Service Intermountain Forest and Range Experiment Station.
- Bruce, D. 1961. *Prism cruising in the western United States and volume tables for use therewith*. Portland, Oregon: Mason, Bruce & Girard, Inc.
- Cancino, J., & Saborowski, J. 2005. Comparison of randomized branch sampling with and without replacement at the first stage. *Silva Fennica*, **39**, 201–216.
- Canfield, R. H. 1941. Application of the line interception method in sampling range vegetation. *Journal of Forestry*, **39**, 34–40.
- Cassel, C.-M., Särndal, C.-E., & Wretman, J. H. 1977. *Foundations of Inference in Survey Sampling*. New York: Wiley.
- Chaudhuri, A., & Stenger, H. 1992. *Survey Sampling: Theory and Methods*. New York: Marcel Dekker.
- Chaudhuri, A., & Vos, J. W. E. 1988. *Unified Theory and Strategies of Survey Sampling*. Amsterdam: North Holland.
- Chen, Z., Bai, Z., & Sinha, B. K. 2003. *Ranked Set Sampling: Theory and Applications*. Lecture Notes in Statistics, 176. New York: Springer-Verlag.
- Christman, M. C. 2000. A review of quadrat-based sampling of rare, geographically clustered populations. *Journal of Agricultural, Biological, and Environmental Statistics*, **5**, 168–201.
- Clapham, A. R. 1932. The form of the observational unit in quantitative ecology. *Journal of Ecology*, **20**, 192–197.
- Clark, III, A., Souter, R. A., & Schlaegel, B. E. 1991. *Stem profile equations for southern tree species*. Research Paper SE-282. Washington, D.C.: USDA Forest Service.
- Clark, S. J., & Perry, J. N. 1994. Small sample estimation of Taylor's power law. *Environmental and Ecological Statistics*, **1**, 287–302.
- Clements, F. E. 1905. *Research Methods in Ecology*. Lincoln, Nebraska (U.S.A): The University Publishing Company.
- Cobby, M. L., Ridout, M. S., Bassett, P. J., & Large, R. V. 1985. An investigation into the use of ranked set sampling on grass and grass-clover swards. *Grass and Forage Science*, **40**, 257–263.
- Cochran, W. G. 1953. *Sampling Techniques*. New York: Wiley.
- Cochran, W. G. 1977. *Sampling Techniques*. New York: Wiley.
- Cochran, W. G., Mosteller, F., & Tukey, J. W. 1954. Principles of sampling. *Journal of the American Statistical Association*, **49**, 13–35.

- Cordy, C. 1993. An extension to the Horvitz-Thompson theorem to point sampling from a continuous population. *Probability and Statistics Letters*, **18**, 353–362.
- Cunia, T. 1979. *Basic Designs for Survey Sampling: Simple, Stratified, Cluster, and Systematic Sampling*. Forest Biometry Monograph Series 3. College of Environmental Science and Forestry, State University of New York, Syracuse, New York, USA.
- Cunia, T., & Briggs, R. D. 1984. Forcing additivity of biomass tables: some empirical results. *Canadian Journal of Forest Research*, **14**, 376–384.
- Cunia, T., & Chevrou, R. B. 1969. Sampling with partial replacement on three or more occasions. *Forest Science*, **15**, 204–224.
- Curtis, R. O., & Marshall, D. D. 2005. *Permanent-plot procedures for silvicultural research*. General Technical Report, PNW-634. Washington, D.C.: USDA Forest Service.
- Dalenius, T. 1957. *Sampling in Sweden: Contributions to the Methods and Theories of Sample Survey Practice*. Stockholm: Almquist and Wiskell.
- Dalenius, T., Hájek, J., & Zubrzycki, S. 1961. On plane sampling and related geometrical problems. Pages 125–150 of: *Proceedings of the 4th Berkeley Symposium on Probability and Mathematical Statistics, Vol. 1*. Berkeley: University of California Press.
- Daubenmire, R. 1959. A canopy-coverage method of vegetational analysis. *North-west Science*, **33**(1), 43–64.
- Davison, A. C., & Hinkley, D. V. 1997. *Bootstrap Methods and Their Application*. Cambridge: Cambridge University Press.
- De Gier, A. 1989. *Woody Biomass for Fuel: Estimating the Supply in Natural Woodlands and Shrublands*. ITC Publication Number 9. Enschede, The Netherlands: International Institute for Aerospace Survey and Earth Sciences (ITC).
- de Gruijter, & ter Braak, C. J. F. 1990. Model-free estimation from spatial samples: a reappraisal of classical sampling theory. *Mathematical Geology*, **22**, 407–415.
- de Vries, P. G. 1973. *A general theory on line intersect sampling, with application to logging residue inventory*. Tech. rept. 73-11. Mededelingen Landbouwhogeschool, Wageningen, The Netherlands.
- de Vries, P. G. 1986. *Sampling Theory for Forest Inventory*. Berlin: Springer-Verlag.
- Delisle, G. P., Woodard, P. M., Titus, S. J., & Johnson, A. F. 1988. Sample size and variability of fuel weight estimates in natural stands of lodgepole pine. *Canadian Journal of Forest Research*, **18**, 649 – 652.
- Dell, T. R., & Clutter, J. L. 1972. Ranked set sampling theory with order statistics background. *Biometrics*, **28**, 545–555.
- Deming, W. E. 1950. *Some Theory of Sampling*. New York: Wiley.
- Deming, W. E., & Stephan, F. F. 1940. On a least squares adjustment of a sample frequency tables when the expected marginal totals are known. *Annals of Mathematical Statistics*, **11**, 427–444.

- Dixon, P. M. 2002. Bootstrap resampling. Pages 212–220 of: *Encyclopedia of Environmetrics, Volume 1*. Chichester: Wiley.
- Ducey, M. J., Jordan, G. J., Gove, J. H., & Valentine, H. T. 2002. A practical modification of horizontal line sampling for snag and cavity tree inventory. *Canadian Journal of Forest Research*, **32**, 1217–1224.
- Ducey, M. J., Gove, J. H., & Valentine, H. T. 2004. A walk-through solution to the boundary overlap problem. *Forest Science*, **50**, 427–435.
- Ducey, M. J., Williams, M. S., Gove, J. H., & Valentine, H. T. 2007. Simultaneous unbiased estimates of multiple downed wood attributes in perpendicular distance sampling. *Submitted to: Canadian Journal of Forest Research*.
- Duncan, G. J., & Kalton, G. 1987. Issues of design and analysis of surveys across time. *International Statistical Review*, **55**, 97–117.
- Efron, B., & Tibshirani, R. 1986. *An Introduction to the Bootstrap*. New York: Chapman and Hall.
- Eriksson, M. 1995a. Compatible and time-additive change component estimators for horizontal-point-sampled data. *Forest Science*, **41**, 796–822.
- Eriksson, M. 1995b. Design-based approaches to horizontal-point sampling. *Forest Science*, **41**, 890–907.
- Evans, M., & Swartz, T. 2000. *Approximating integrals via Monte Carlo and Deterministic Methods*. Oxford: Oxford University Press.
- FAO. 1990. *Forest Resources Assessment 1990*. FAO Forestry paper 124. Rome: United Nations Food and Agriculture Organization.
- FHM. 1998. *Forest Health Monitoring Methods Guide*. Research Triangle Park, North Carolina: USDA Forest Service, National Forest Health Monitoring Program.
- Flewelling, J., & Iles, K. 2004. Area-independent sampling for total basal area. *Forest Science*, **50**(4), 512–517.
- Fowler, G. W., & Hauke, D. 1979. *A distribution-free method for interval estimation and sample size determination*. Resource Inventory Notes BLM-19. Denver, Colorado: Bureau of Land Management, U. S. Department of Interior.
- Frayser, W. E. 1978. *Stratification in double sampling – The easy way out may sometimes be the best way*. Resource Inventory Notes BLM-10. Denver, Colorado: Bureau of Land Management, U. S. Department of Interior.
- Freese, F. 1961. Relation of plot size to variability. *Journal of Forestry*, **59**, 679.
- Fuller, W. A. 1990. Analysis of repeated surveys. *Statistical Methodology*, **16**(2), 167–180.
- Fuller, W. A. 1995. Estimation in the presence of measurement error. *International Statistical Review*, **63**, 121–147.
- Fuller, W. A. 1999. Environmental surveys over time. *Journal of Agricultural, Biological, and Environmental Statistics*, **4**(4), 331–345.

- Fuller, W. A. 2002. Regression estimation for survey samples. *Survey Methodology*, **28**, 5–23.
- Furness, G. O. 1976. The dispersal, age-structure and natural enemies of the long-tailed mealy bug, *Pseudococcus longispinus* (Targoni-Tozzetti) in relation to sampling and control. *Australian Journal of Zoology*, **24**, 237–247.
- Furnival, G. M., Gregoire, T. G., & Grosenbaugh, L. R. 1987. Adjusted inclusion probabilities with 3P sampling. *Forest Science*, **33**(3), 617–631.
- Gaffrey, D., & Saborowski, J. 1999. RBS, ein mehrstufiges Inventurverfahren zur Schätzung von Baummerkmalen I. Schätzung von Nadel- und Asttrockenmassen bei 66-jährigen Douglasien. *Allg. Forst. u.J.-Ztg.*, **170**, 177–183.
- Gilbert, R. O., & Eberhardt, L. L. 1976. An evaluation of double sampling for estimating plutonium inventory in surface soil. Pages 157–163 of: Cushing, C. E. et. al. (ed), *Radioecology and energy resources : proceedings of the Fourth National Symposium on Radioecology*. The Ecological Society of America, Special Publication No. 1. Stroudsburg, Pennsylvania: Dowden, Hutchinson and Ross, Inc.
- Godambe, V. P. 1955. A unified theory of sampling from finite populations. *Journal of the Royal Statistical Society, Series B*, **17**, 269–278.
- Good, N. M., Paterson, M., Brack, C., & Mengersen, K. 2001. Estimating tree component biomass using variable probability sampling methods. *Journal of Agricultural, Biological and Environmental Statistics*, **6**, 258–267.
- Goodman, L. A. 1949. On the estimation of the number of classes in a population. *Annals of Mathematical Statistics*, **20**, 572–579.
- Goodman, L. A. 1960. On the exact variance of products. *Journal of the American Statistical Association*, **55**, 708–713.
- Gove, J. H., Ducey, M. J., Ståhl, G., & Ringvall, A. 1999a. Point relascope sampling: a new way to assess downed coarse woody debris. *Journal of Forestry*, **99**, 4–11.
- Gove, J. H., Ringvall, A., Ståhl, G., & Ducey, M. J. 1999b. Point relascope sampling for downed coarse woody debris. *Canadian Journal of Forest Research*, **29**, 1718–1726.
- Gray, H. R. 1943. Volume measurement of forest crops. *Australian Forestry*, **7**, 48–74.
- Graybill, F. A., & Deal, R. B. 1959. Combining unbiased estimators. *Biometrics*, **15**, 543–550.
- Greenhill, A. H. 1881. Determination of the greatest height consistent with stability that a vertical pole or mast can be made, and the greatest height to which a tree of given proportions can grow. *Proceedings of the Cambridge Philosophical Society*, **4**, 65–73.
- Gregoire, T. G. 1982. The unbiasedness of the mirage correction procedure for boundary overlap. *Forest Science*, **28**(3), 504–508.
- Gregoire, T. G. 1984. The jackknife: an introduction with applications in forestry data analysis. *Canadian Journal of Forest Research*, **14**, 493–497.

- Gregoire, T. G. 1992. Analytical derivation of the variance of the modified point-list sampling estimator. Pages 404–415 of: Wood, G., & Turner, B. (eds), *Integrating Forest Information Over Space and Time*. Proceedings of IUFROM S4.01 & S4.02 conference. Canberra, Australia: ANUTECH Pty. Ltd.
- Gregoire, T. G. 1993. Estimation of growth from successive surveys. *Forest Ecology and Management*, **56**, 267–278.
- Gregoire, T. G. 1998. Design-based and model-based inference in survey sampling: appreciating the difference. *Canadian Journal of Forest Research*, **28**, 1429–1447.
- Gregoire, T. G., & Monkevich, N. S. 1994. The reflection method of line intercept sampling to eliminate boundary bias. *Environmental and Ecological Statistics*, **1**, 219–226.
- Gregoire, T. G., & Schabenberger, O. S. 1999. Sampling skewed biological populations: behavior of confidence intervals for the population total. *Ecology*, **80**, 1056–1065.
- Gregoire, T. G., & Scott, C. T. 1990. Sampling at the stand boundary: a comparison of the statistical performance among eight methods. Pages 78–85 of: Burkhardt, H. E., Bonnor, G. M., & Lowe, J. J. (eds), *Research in Forest Inventory, Monitoring, Growth and Yield*. Publ. FWS-3-90. School of Forestry and Wildlife Resources, Virginia Polytechnic Institute and State University, Blacksburg, Virginia (USA).
- Gregoire, T. G., & Scott, C. T. 2003. Altered selection probabilities caused by avoiding the edge in field surveys. *Journal of Agricultural, Biological and Environmental Statistics*, **8**, 36–47.
- Gregoire, T. G., & Valentine, H. T. 1999. Composite and calibration estimation following 3P sampling. *Forest Science*, **45**(2), 179–185.
- Gregoire, T. G., & Valentine, H. T. 2003. Line intersect sampling: ell-shaped transects and multiple intersections. *Environmental and Ecological Statistics*, **10**, 263–279.
- Gregoire, T. G., Valentine, H. T., & Furnival, G. M. 1986. Estimation of bole volume by importance sampling. *Canadian Journal of Forest Research*, **16**, 554–557.
- Gregoire, T. G., Valentine, H. T., & Furnival, G. M. 1993. Two-stage and three-stage sampling strategies to estimate aggregate bole volume in the forest. Pages 201–211 of: Nyssonen, A., Poso, S., & Rautala, J. (eds), *Ilvessalo Symposium on National Forest Inventories*. Research Paper 444. Helsinki, Finland: The Finnish Forest Research Institute.
- Gregoire, T. G., Valentine, H. T., & Furnival, G. M. 1995. Sampling methods to estimate foliage and other characteristics of individual trees. *Ecology*, **76**, 1181–1194.
- Grosenbaugh, L. R. 1952. Plotless, timber estimates – new, fast, easy. *Journal of Forestry*, **50**(1), 32–37.
- Grosenbaugh, L. R. 1958. *Point-sampling and line-intercept sampling: probability theory, geometric implications, synthesis*. Southern Forest Experiment Station Occasional Paper SO-160. Washington, D.C.: USDA Forest Service.

- Grosenbaugh, L. R. 1964. Some suggestions for better sample-tree measurement. Pages 36–42 of: anon. (ed), *Proceedings, Society of American Foresters*. Boston, Massachusetts: Society of American Foresters.
- Grosenbaugh, L. R. 1965. *THREE-PEE SAMPLING THEORY and program 'THRP' for computer generation of selection criteria*. Research Paper, PSW-21. Berkeley, California: USDA Forest Service.
- Grosenbaugh, L. R. 1971. *STX 1-11-71 for Dendrometry of Multistage 3P Samples*. USDA Forest Service.
- Grosenbaugh, L. R. 1976. Approximate sampling variance of adjusted 3P estimates. *Forest Science*, **22**, 173–176.
- Gupta, P. C. 1972. A Monte Carlo integration approach to Horvitz-Thompson estimation in replicated environmental designs. *Australian Journal of Statistics*, **14**, 123–128.
- Hájek, J. 1958. Some contributions to the theory of probability sampling. *Bulletin of the International Statistical Institute*, **36**, 127–134.
- Hájek, J. 1964. Asymptotic theory of rejective sampling with varying probabilities from a finite population. *Annals of Mathematical Statistics*, **35**(4), 1491–1523.
- Hájek, J. 1981. *Sampling from a finite population*. New York: Marcel Dekker.
- Halls, L. S., & Dell, T. R. 1966. Trial of ranked set sampling for forage yields. *Forest Science*, **12**, 22–26.
- Hammersley, J. M., & Handscomb, D. C. 1979. *Monte Carlo Methods*. London: Chapman and Hall.
- Hammersley, J. M., & Morton, K. W. 1956. A new Monte Carlo technique: antithetic variates. *Proceedings of the Cambridge Philosophical Society*, **52**, 449–475.
- Hansen, M. H., & Hurwitz, W. N. 1943. On the theory of sampling from finite populations. *Annals of Mathematical Statistics*, **14**, 333–362.
- Hansen, M. H., Hurwitz, W. N., & Madow, W. G. 1953. *Sample Survey Methods and Theory, Volume 1*. New York: Wiley.
- Hansen, M. R., Dalenius, T., & Tepping, B. J. 1985. The development of sample surveys of finite populations. Pages 327–354 of: Atkinson, A. C., & Feinberg, S. E. (eds), *A Celebration of Statistics*. New York: Springer-Verlag.
- Hanson, H. C. 1934. A comparison of methods of botanical analysis of the native prairie in western North Dakota. *Journal of Agricultural Research*, **49**(9), 815–842.
- Hartley, H. O., & Rao, J. N. K. 1962. Sampling with unequal probabilities and without replacement. *Annals of Mathematical Statistics*, **33**, 350–374.
- Holt, D., & Smith, T. M. F. 1979. Post stratification. *Journal of the Royal Statistical Society, Series A*, **142**, 33–46.
- Horvitz, D. G., & Thompson, D. J. 1952. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, **47**, 663–685.

- Howard, J. O., & Ward, F. R. 1972. *Measurement of Logging Residue – Alternative Applications of the Line Intersect Method*. Research Note PNW-183. Washington, D.C.: USDA Forest Service.
- Iles, K. 1989. Subsampling for VBAR. *Inventory Cruising News*, **8**, 2–3.
- Iles, K. 2003. *A Sampler of Inventory Topics*. Nanaimo, BC, Canada: Kim Iles & Associates.
- Jenkins, J. C., Chojnacky, D. C., Heath, L. S., & Birdsey, R. A. 2004. *Comprehensive database of diameter-based biomass regressions for North American tree species*. General Technical Report, NE-319. Washington, D.C.: USDA Forest Service.
- Jessen, R. J. 1942. *Statistical Investigation of a sample survey for obtaining farm facts*. Research Bulletin 304. Ames, Iowa: Iowa Agricultural Experiment Station.
- Jessen, R. J. 1955. Determining the fruit count on a tree by randomized branch sampling. *Biometrics*, **11**, 99–109.
- Jessen, R. J. 1978. *Statistical Survey Techniques*. New York: Wiley.
- Johnson, F. A. 1961. *Standard error of estimated average timber volume per acre under point sampling when trees are measured for volume on a subsample of all points*. Research Note, PNW-201. Portland, Oregon: USDA Forest Service.
- Johnson, F. A., & Hixon, H. J. 1952. The most efficient size and shape plot to use for cruising in old-growth Douglas-fir timber. *Journal of Forestry*, **50**(1), 17–20.
- Jones, H. L. 1958. Inadmissible samples and confidence limits. *Journal of the American Statistical Association*, **53**, 482–490.
- Jönrup, H., & Rennermalm, B. 1976. Regression analysis in samples from finite populations. *Scandinavian Journal of Statistics*, **3**, 33–36.
- Kaiser, L. 1983. Unbiased estimation in line-intercept sampling. *Biometrics*, **39**, 965–976.
- Kaur, A., Patil, G. P., Sinha, A. K., & Tallie, C. 1995. Ranked set sampling: an annotated bibliography. *Environmental and Ecological Statistics*, **2**, 25–54.
- Koop, J. C. 1990. Systematic sampling of two-dimensional surfaces and related problems. *Communications in Statistics–Theory and Methods*, **19**, 1701–1759.
- Kraft, K. M., Johnson, D. H., Samuleson, J. M., & Allen, S. H. 1995. Using known populations of pronghorn to evaluate sampling plans and estimators. *Journal of Wildlife Management*, **59**, 129–137.
- Kruskal, W. H., & Mosteller, F. 1979a. Representative sampling, I. Nonscientific literature. *International Statistical Review*, **47**, 13–24.
- Kruskal, W. H., & Mosteller, F. 1979b. Representative sampling, II. Scientific literature, excluding statistics. *International Statistical Review*, **47**, 111–127.
- Kruskal, W. H., & Mosteller, F. 1979c. Representative sampling, III. The current statistical literature. *International Statistical Review*, **47**, 245–265.
- Kulow, D. R. 1966. Comparison of forest sampling designs. *Journal of Forestry*, **64**, 469–474.

- Lahiri, D. B. 1951. A method of sample selection providing unbiased ratio estimates. *Bulletin the International Statistical Institute*, **33**, 133–140.
- Lazzerini, M. 1901. Un'applicazione del calcolo della probabilità alla ricerca di un valore approssimato di π . *Periodico di Matematiche*, **4**, 140–143.
- Leemis, L. M., & Trivedi, K. S. 1996. A comparison of approximate interval estimators for the Bernoulli parameter. *The American Statistician*, **50**, 63–68.
- Lohr, S. L. 1999. *Sampling: Design and Analysis*. Pacific Grove, California: Duxbury Press.
- Lucas, H. A., & Seber, G. A. F. 1977. Estimating coverage and particle density using the line intercept method. *Biometrika*, **64**, 618–622.
- MacEachern, S. N., Öztürk, Ö., & Wolfe, D. A. 2002. A new ranked set sample estimator of variance. *Journal of the Royal Statistical Society, B*, **64**, Part 2, 177–188.
- MacLean, C. D. 1972a. Improving inventory volume estimates by double sampling on aerial photographs. *Journal of Forestry*, **70**, 748–749.
- MacLean, C. D. 1972b. *Photo stratification improves Northwest timber volume estimates*. Research Paper, PNW-150. Washington, D.C.: USDA Forest Service.
- Magnussen, S., Kurz, W., & Leckie, D. P. 2005. Adaptive cluster sampling for estimation of deforestation rates. *European Journal of Forest Research*, **124**, 207–220.
- Mahalanobis, P. C. 1946. Recent experiments in statistical sampling in the Indian Statistical Institute. *Journal of the Royal Statistical Society, A*, **109**, 326–378.
- Mandallaz, D. 1991. *A Unified Approach to Sampling Theory for Forest Inventory Based on Infinite Population and Superpopulation Models*. Zurich: Chair of Forestry Inventory and Planning, Swiss Federal Institute of Technology (ETH).
- Marshall, A. W. 1956. The use of multi-stage sampling schemes in Monte Carlo computations. Pages 123–140 of: Meyer, M. A. (ed), *Symposium on Monte Carlo Methods*. New York: Wiley.
- Marshall, D. D., Iles, K., & Bell, J. F. 2004. Using a large-angle gauge to select trees for measurement in variable plot sampling. *Canadian Journal of Forest Research*, **34**, 840–845.
- Martin, G. L. 1982. A method for estimating ingrowth on permanent horizontal sample points. *Forest Science*, **28**, 110–114.
- Martin, W. L., Sharik, T. L., Oderwald, R. G., & Smith, D. W. 1980. *Evaluation of Ranked Set Sampling for Estimating Shrub Phytomass in Appalachian Oak Forests*. Publication No. FWS-4-80. Blacksburg: School of Forestry and Wildlife Resources, Virginia Polytechnic Institute and State University.
- Masuyama, M. 1954. On the error in crop cutting experiment due to the bias on the border of the grid. *Sankhyā*, **14**(3), 181–186.
- Matérn, B. 1956. On the geometry of the cross-section of a stem. *Meddelanden Från Statens Skogsforskningsinstitut*, **Band 46 Nr(11)**, 1–28.

- Matérn, B. 1964. A method of determining the total length of roads by means of a line survey. *Studia Forestalia Suecica*, **18**, 68–70.
- Matney, T. G., & Parker, R. C. 1991. Stand and stock tables from double-point samples. *Forest Science*, **37**, 1605–1613.
- McIntyre, G. A. 1952. A method for unbiased selective sampling, using ranked sets. *Australian Journal of Agricultural Research*, **3**, 385–390.
- McLeod, A. I., & Bellhouse, D. R. 1983. A convenient algorithm for drawing a simple random sample. *Applied Statistics*, **32**, 182–184.
- Meier, P. 1953. Variance of a weighted mean. *Biometrics*, **9**, 59–73.
- Mendenhall, W., & Schaeffer, R. L. 1973. *Mathematical Statistics with Applications*. North Scituate, Massachusetts: Duxbury Press.
- Mesavage, C., & Grosenbaugh, L. R. 1956. Efficiency of several cruising designs on small tracts in north Arkansas. *Journal of Forestry*, **54**(9), 569–576.
- Meyer, H. A. 1948. Cruising by narrow strips. *Pennsylvania State Forestry School Research Paper 12*.
- Meyer, H. A. 1958. The calculation of the sampling error of a cruise from the mean square successive difference. *Journal of Forestry*, **54**, 341.
- Monkevich, N. S. 1994. *Assessing petrified wood changes in Petrified Wood National Park*. M.Phil. thesis, Virginia Polytechnic Institute and State University.
- Murray, C. D. 1927. A relationship between circumference and weight in trees and its bearing on branching angles. *Journal of General Physiology*, **10**, 725–739.
- Murthy, M. N. 1964. Product method of estimation. *Sankhyā, Series A*, **26**, 69–74.
- Murthy, M. N. 1967. *Sampling: Theory and Methods*. Calcutta: Statistical Publishing Association.
- Muttlak, H. A., & Sadooghi-Alvandi, S. M. 1993. A note on the line intercept sampling method. *Biometrics*, **49**, 1209–1215.
- Nelson, R., & Gregoire, T. G. 1994. Two-stage forest sampling: a comparison of three procedures to estimate aggregate volume. *Forest Science*, **40**(2), 247–266.
- Nelson, R., Short, A., & Valenti, M. 2004. Measuring biomass and carbon in Delaware using an airborne profiling LIDAR. *Scandinavian Journal of Forest Research*, **19**(6), 500–511.
- Newman, E. I. 1966. A method of estimating the total length of root in a sample. *Journal of Applied Ecology*, **3**(1), 139–145.
- Newton, C. M., Cunia, T., & Bickford, C. A. 1974. Multivariate estimators for sampling with partial replacement on two occasions. *Forest Science*, **20**, 106–116.
- Neyman, J. 1934. On the two different aspects of the representative method: the method of stratified sampling and the method of purposive selection. *Journal of the Royal Statistical Society*, **97**, 558–606.
- Neyman, J. 1938. Contribution to the theory of sampling human populations. *Journal of the American Statistical Association*, **33**, 101–116.

- Nusser, S. M., & Goebel, J. J. 1997. The National Resources Inventory: a long-term multi-resource monitoring programme. *Environmental and Ecological Statistics*, **4**, 181–204.
- Nusser, S. M., Breidt, F. J., & Fuller, W. A. 1998. Design and estimation for investigating the dynamics of natural resources. *Ecological Applications*, **8**, 234–245.
- Oderwald, R. G. 1993. Stratification gains in cruising: proportional allocation. *Southern Journal of Applied Forestry*, **17**, 96–99.
- Oderwald, R. G. 1994. Stock and stand tables for point, double sampling with a ratio of means estimator. *Canadian Journal of Forest Research*, **24**, 2350–2352.
- Oderwald, R. G., & Jones, E. 1992. Sample sizes for point, double sampling. *Canadian Journal of Forest Research*, **22**, 980–983.
- Olsen, A. R., Sedransk, J., Edwards, D., Gotway, C. A., Liggett, W., Rathbun, S., Reckhow, K. H., & Young, L. J. 1999. Statistical issues for monitoring ecological and natural resources in the United States. *Environmental Monitoring and Assessment*, **54**(1), 1–45.
- Osborne, J. G. 1942. Sampling errors of systematic and random surveys of cover-type areas. *Journal of the American Statistical Association*, **37**, 256–264.
- Overton, W. S., & Stehman, S. V. 1993. Properties of designs for sampling continuous spatial resources from a triangular grid. *Communications in Statistics—Theory and Methods*, **22**, 2641–2660.
- Palley, M. N., & Horwitz, L. G. 1961. Properties of some random and systematic point sampling estimators. *Forest Science*, **7**, 52–65.
- Patil, G. P., Sinha, A. K., & Tallie, C. 1994. Ranked set sampling. Pages 167–200 of: Patil, G. P., & Rao, C. R. (eds), *Handbook of Statistics, Volume 12: Environmental Statistics*. Amsterdam: North Holland.
- Patterson, H. D. 1950. Sampling on successive occasions with partial replacement of units. *Journal of the Royal Statistical Society, B*, **12**(2), 241–255.
- Penner, M., & Otukol, S. 1998. Boundary correction for variable radius plots—simulation results. Pages 148–157 of: Hansen, M., & Burk, T. (eds), *Integrated Tools for Natural Resources Inventories in the 21st Century*. St. Paul, Minnesota: USDA Forest Service.
- Perlman, M. D., & Wichura, M. J. 1975. Sharpening Buffon's Needle. *The American Statistician*, **29**(4), 157–163.
- Pfeffermann, D., & Krieger, A. M. 1991. Poststratification using regression estimators when information on strata means and sizes is missing. *Biometrika*, **78**, 409–419.
- Phillipi, T. 2005. Adaptive cluster sampling for estimation of abundances within local populations of low-abundance plants. *Ecology*, **86**, 1091–1100.
- Pinkham, R. S. 1987. An efficient algorithm for drawing a simple random sample. *Applied Statistics*, **36**, 370–372.

- Ponce-Hernandez, R. 2004. *Assessing carbon stocks and modelling win-win scenarios of carbon sequestration through land-use changes*. Rome: Food and Agriculture Organization, United Nations.
- Pontius, J. S. 1996. Forest spatial surveys using the Rao-Hartley-Cochran sampling design. In: *Spatial Accuracy Assessment in Natural Resources and Environmental Sciences: Second International Symposium*. General Technical Report RM-GTR-277. Washington, D.C.: USDA Forest Service.
- Pontius, J. S. 1998. Estimation of the mean in line intercept sampling. *Environmental and ecological statistics*, **5**, 371–379.
- Quenouille, M. H. 1949. Problems in plane sampling. *Annals of Mathematical Statistics*, **20**, 355–375.
- Quenouille, M. H. 1956. Notes on bias in estimation. *Biometrika*, **43**, 353–360.
- Radtke, P. J., & Bolstad, P. V. 2001. Laser point-quadrat sampling for estimating foliage-height profiles in broad-leaved forests. *Canadian Journal of Forest Research*, **31**, 410–418.
- Raj, D. 1968. *Sampling Theory*. New York: McGraw-Hill.
- Ramaley, J. F. 1969. Buffon's noodle problem. *The American Mathematical Monthly*, **76**(8), 916–918.
- Rao, J. N. K. 2003. *Small Area Estimation*. Hoboken, New Jersey: Wiley-Interscience.
- Rao, J. N. K., & Ramachandran, V. 1974. Comparison of the separate and combined ratio estimators. *Sankhyā*, **36**, 151–156.
- Rao, J. N. K., Hartley, H. O., & Cochran, W. G. 1962. On a simple procedure of unequal probability sampling without replacement. *Journal of the Royal Statistical Society, Series B*, **24**, 482–491.
- Rao, P. S. R. S. 1988. Ratio and regression estimators. Pages 449–468 of: Krishnaiah, P. R., & Rao, C. R. (eds), *Handbook of Statistics 6: Sampling*. Amsterdam: North Holland.
- Raulier, F., Bernier, P. Y., Ung, C.-H., & Boutin, R. 2002. Structural differences and functional similarities between two sugar maple (*Acer saccharum*) stands. *Tree Physiology*, **22**, 1147–1156.
- Rennolls, K. 1989. *Design of the Census of Woodlands and Trees 1979-82*. Edinburgh: Occasional Paper 18, Forestry Commission.
- Ringvall, A., & Kruys, N. 2005. Sampling of sparse species with probability proportional to prediction. *Environmental Monitoring and Assessment*, **104**, 131–146.
- Risebrough, R. W. 1972. Effects of pollutants upon animals other than man. In: *Proceedings of the 6th Berkeley Symposium on Mathematics and Statistics, VI*. Berkeley: University of California Press.
- Robson, D. S. 1957. Applications of multivariate polykays to the theory of unbiased ratio-type estimation. *Journal of the American Statistical Association*, **52**, 511–522.

- Roesch, Jr., F. A., Green, E. J., & Scott, C. T. 1989. New compatible estimators for survivor growth and ingrowth from remeasured horizontal point samples. *Forest Science*, **35**, 281–293.
- Roesch, F. A., Jr. 1993. Adaptive cluster sampling for forest inventories. *Forest Science*, **39**, 655–669.
- Rosiwal, A. 1898. Ueber geometrische Gesteinsanalysen. Ein einfacher Weg zur ziffremassigen Foxtstellung des Quantitätsverhältnissos der Mineralbestandtheile gemengter Geneine. *Verhandlungen*, **5(6)**, 143–174.
- Rubinstein, R. Y. 1981. *Simulation and the Monte Carlo Method*. New York: Wiley.
- Rudis, V. A. 1991. *Wildlife Habitat, Range, Recreation, Hydrology, and Related Research Using Forest Inventory and Analysis Surveys: A 12-Year Compendium*. General Technical Report SO-84. Washington, D.C.: USDA Forest Service.
- Särndal, C.-E. 1978. Design-based and model-based inference in survey sampling. *Scandinavian Journal of Statistics*, **5**, 27–52.
- Särndal, C.-E. 1982. Implications of survey design for generalized regression estimation of linear functions. *Journal of Statistical Planning and Inference*, **7**, 155–170.
- Särndal, C.-E. 1996. Efficient estimators with simple variance in unequal probability sampling. *Journal of the American Statistical Association*, **91**, 1289–1300.
- Särndal, C.-E., Swensson, B., & Wretman, J. 1992. *Model Assisted Survey Sampling*. New York: Springer-Verlag.
- Schabenberger, O. S., & Gregoire, T. G. 1994. Competitors to genuine π ps designs: a comparison. *Survey Methodology*, **20**, 185–192.
- Schmid, V. P. 1969. Stichproben am waldtrand (Sample plots at forest stand margins). *Schweizerische Anstalt für das Forstliche Versuchswesen, Mitteilungen*, **45(3)**, 234–303.
- Schreuder, H. T., Sedransk, J., & Ware, K. D. 1968. 3-P sampling and some alternatives, I. *Forest Science*, **14**, 429–454.
- Schreuder, H. T., Gregoire, T. G., & Wood, G. B. 1993. *Sampling Methods for Multiresource Forest Inventory*. New York: Wiley.
- Schumacher, F. X., & Bull, H. 1932. Determination of the errors of estimate of a forest survey, with special reference to the bottom-land hardwood forest region. *Journal of Agricultural Research*, **45(12)**, 741–756.
- Scott, C. T. 1984. A new look at sampling with partial replacement. *Forest Science*, **30**, 157–166.
- Scott, C. T., & Köhl, M. 1994. Sampling with partial replacement and stratification. *Forest Science*, **40**, 30–46.
- Scott, C. T., Köhl, & Schnellbächer, H. J. 1999. A comparison of periodic and annual forest inventories. *Forest Science*, **45(3)**, 433–451.
- Seber, G. A. F. 1979. Transects of random length. Pages 183–192 of: McCormack, R. M., Patil, G. P., & Robson, D. S. (eds), *Sampling Biological Populations*. Fairfield, Maryland, USA: International Cooperative Publishing House.

- Seber, G. A. F. 1982. *The Estimation of Animal Abundance and Related Parameters*. London: Charles Griffin & Company Ltd.
- Sengupta, J. M. 1954. Some experiments with different types of area sampling for winter paddy in Giridh, Bihar: 1945. *Sankhyā*, **13**, 235–240.
- Shinozaki, K., Yoda, K., Hozumi, K., & Kira, T. 1964. A quantitative analysis of plant form: the pipe model theory. I. Basic analysis. *Japanese Journal of Ecology*, **14**, 97–105.
- Singh, H., & Espejo, M. R. 2003. On linear regression and ratio-product estimation of finite population mean. *The Statistician*, **52**, 59–67.
- Singh, S., & Horn, S. 1998. An alternative estimator for multicharacter surveys. *Metrika*, **48**, 99–107.
- Sitter, R. R. 1992. A resampling procedure for complex survey data. *Journal of the American Statistical Association*, **87**, 755–765.
- Skidmore, A. K., & Turner, B. J. 1992. Map accuracy assessment using line intersect sampling. *Photogrammetric Engineering and Remote Sensing*, **58(10)**, 1453–1457.
- Smith, C., & Guttman, L. 1953. Measurement of internal boundaries in three-dimensional structures by random sampling. *Journal of Metals*, **5(1)**, 81–87.
- Smith, D. R., Villela, R. F., & Lamarié, D. P. 2003. Application of adaptive cluster sampling to low-density populations of freshwater mussels. *Environmental and Ecological Statistics*, **10**, 7–15.
- Smith, H. F. 1938. An empirical law describing heterogeneity in the yields of agricultural crops. *Journal of Agricultural Sciences*, **28**, 1–23.
- Smith, S. J., & Gavaris, S. 1993. Improving the precision of abundance estimates of eastern Scotian shelf Atlantic cod from bottom trawl surveys. *North American Journal of Fisheries Management*, **13**, 35–47.
- Smith, T. M. F. 1991. Post stratification. *The Statistician*, **40**, 315–323.
- Smith, V. G., & Aird, P. L. 1975. *Canadian Forest Inventory Methods*. Toronto: University of Toronto Press.
- Spetich, M. A., & Parker, G. R. 1998. Plot size recommendations for biomass estimation in midwestern old-growth forest. *Northern Journal of Applied Forestry*, **15(4)**, 165–168.
- Stage, A. R., & Rennie, J. C. 1994. Fixed-radius plots or variable-radius plots? *Journal of Forestry*, **92(12)**, 20–24.
- Ståhl, G. 1997. Transect relascope sampling for assessing coarse woody debris: the case for a $\pi/2$ relascope angle. *Scandinavian Journal of Forest Research*, **12**, 375–381.
- Ståhl, G. 1998. Transect relascope sampling—a method for the quantification of coarse woody debris. *Forest Science*, **44**, 58–63.
- Stehman, S. V., & Overton, W. S. 2002. Environmental Sampling. Pages 1914–1937 of: *Encyclopedia of Environmetrics, Volume 4*. Chichester: Wiley.

- Stehman, S. V., & Salzer, D. W. 2000. Estimating density from surveys employing unequal-area belt transects. *Wetlands*, **20**, 512–519.
- Stehman, S. V., Sohl, T. L., & Loveland, T. R. 2003. Statistical sampling to characterize recent United States land-cover change. *Remote Sensing of Environment*, **20**, 512–519.
- Stevens, Jr., D. L. 1997. Variable density grid-based sampling designs for continuous spatial populations. *Environmetrics*, **8**, 167–195.
- Stevens, Jr., D. L., & Olsen, A. R. 1991. Statistical issues in environmental monitoring and assessment. Pages 76–85 of: *Proceedings of the Section on Statistics and the Environment*. Alexandria, Virginia: American Statistical Association.
- Stokes, S. L. 1980. Estimation of variance using judgment ordered rank set samples. *Biometrics*, **36**, 35–42.
- Strand, L. 1957. “Relaskopisk” hoyde- og kubikkmassebestemmelse. *Norsk Skogbruk*, **3**, 535–538.
- Stuart, A. 1962. *Basic Ideas of Scientific Sampling*. New York: Hafner Press.
- Sudakar, K. 1978. A note of circular systematic sampling design. *Sankhyā, Series C*, **40**, 72–73.
- Sukhatme, P. V. 1947. The problem of plot size in large-scale yield surveys. *Journal of the American Statistical Association*, **42**, 297–310.
- Sukhatme, P. V., & Sukhatme, B. V. 1970. *Sampling Theory of Surveys with Applications*. Second ed. Ames: Iowa State University Press.
- Sukhatme, P. V., Sukhatme, B. V., Sukhatme, S., & Asok, C. 1984. *Sampling Theory of Surveys with Applications*. Third ed. Ames: Iowa State University Press.
- Sunter, A. 1986. Solutions to the problem of unequal probability sampling without replacement. *International Statistical Review*, **54**, 33–50.
- Sunter, A. 1989. Updating size measures in a PPSWOR design. *Survey methodology*, **15**, 253–260.
- Swindel, B., & Yandle, D. O. 1972. Allocation in stratified sampling as a game. *Journal of the American Statistical Association*, **67**, 684–686.
- Takahasi, K., & Wakimoto, K. 1968. On unbiased estimates of the population mean based on the sample stratified by means of ordering. *Annals of the Institute of Statistical Mathematics*, **20**, 1–31.
- Talvitie, M., Leino, O., & Holopainen, M. 2006. Inventory of sparse forest populations using adaptive cluster sampling. *Silva Fennica*, **40**, 101–108.
- Taylor, L. R. 1971. Aggregation, variation, and the mean. *Nature*, **189**, 2732–735.
- Thompson, S. K. 2002. *Sampling*. Second ed. New York: Wiley.
- Thompson, S. K., & Seber, G. A. 1996. *Adaptive Sampling*. New York: Wiley.
- Tillé, Y. 1998. Estimation in surveys using conditional inclusion probabilities: simple random sampling. *International Statistical Review*, **66(3)**, 303–322.

- Tschuprow, A. A. 1922. On the mathematical expectation of the moments of frequency distributions in the case of correlated observations. *Metron*, **2**, 461–493, 646–683.
- Uspensky, J. V. 1937. *Introduction to Mathematical Probability*. New York: McGraw-Hill Book Company, Inc.
- Valentine, H. T., & Hilton, S. J. 1977. Sampling oak foliage by the randomized-branch method. *Canadian Journal of Forest Research*, **7**, 295–298.
- Valentine, H. T., Tritton, L. M., & Furnival, G. M. 1984. Subsampling trees for biomass, volume, or mineral content. *Forest Science*, **30**, 673–681.
- Valentine, H. T., Gregoire, T. G., & Furnival, G. M. 1987. Unbiased estimation of total tree weight by three stage sampling with probability proportional to size. Pages 129–132 of: Wharton, E. H., & Cunia, T. (eds), *Tree Biomass Regression Functions and their Contribution to the Error of Forest Inventory Estimates*. General Technical Report NE-GTR-117. Broomall, Pennsylvania: USDA Forest Service.
- Valentine, H. T., Bealle, C., & Gregoire, T. G. 1992. Comparing vertical and horizontal modes of importance and control-variate sampling for bole volume. *Forest Science*, **38**, 160–172.
- Valentine, H. T., Baldwin, Jr., V. C., Gregoire, T. G., & Burkhart, H. E. 1994. Surrogates for foliar dry matter in loblolly pine. *Forest Science*, **40**, 576–585.
- Valentine, H. T., Gove, J. H., & Gregoire, T. G. 2001. Monte Carlo approaches to sampling forested tracts with lines or points. *Canadian Journal of Forest Research*, **31**, 1410–1424.
- Valentine, H. T., Ducey, M. J., Gove, J. H., Lanz, A., & Affleck, D. L. R. 2006. Corrections for cluster plot slop. *Forest Science*, **52**, 55–66.
- Valiant, R. 1990. Comparisons of variance estimators in stratified random and systematic sampling. *Journal of Official Statistics*, **6**, 115–131.
- Van Deusen, P. C. 1990. Critical height versus importance sampling for log volume: does critical height prevail? *Forest Science*, **36**, 930–938.
- Van Deusen, P. C., & Baldwin, Jr., V.C. 1993. Sampling and predicting tree dry weight. *Canadian Journal of Forest Research*, **23**, 1826–1829.
- Van Deusen, P. C., Dell, T. R., & Thomas, C. E. 1986. Volume growth estimation from permanent horizontal points. *Forest Science*, **32**, 415–422.
- Van Hooser, D. D. 1973. *Midcycle evaluation of Mississippi timber resources*. Resource Bulletin, SO-44. New Orleans, Louisiana: USDA Forest Service.
- van Wagner, C. E. 1968. The line intersect method in forest fuels sampling. *Forest Science*, **14**(1), 267–276.
- Ware, K. D. 1960. *Optimum regression sampling design for sampling of forest populations on successive occasions*. Tech. rept. Yale University.
- Ware, K. D., & Cunia, T. 1962. Continuous forest inventory with partial replacement of samples. *Forest Science Monograph*.

- Warren, W. G., & Olsen, P. F. 1964. A line intersect technique for assessing logging waste. *Forest Science*, **10**, 267–276.
- Watson, G. S. 1978. Characteristics statistical problems of stochastic geometry. Pages 215–234 of: Miles, R. E., & Serra, J. (eds), *Geometrical Probability and Biological Structures: Buffon's 200th Anniversary*. Lecture Notes in Biomathematics, no. 23. Springer-Verlag.
- West, G. B., Brown, J. H., & Enquist, B. J. 1999. A general model for the structure and allometry of plant vascular systems. *Nature*, **400**, 664–667.
- Wiegert, R. G. 1962. The selection of an optimum quadrat size for sampling the standing crop of grasses and forbs. *Ecology*, **43**(1), 125–129.
- Wilk, S. J., Morse, W. W., Ralph, D. E., & Azarovitz, T. R. 1977. *Fishes and associated environmental data collected in New York Bight, June 1974–June 1975*. NOAA Technical Report NMFS SSRF-716. Washington, D.C.: U.S. Department of Commerce.
- Williams, M. S., & Gove, J. H. 2003. Perpendicular distance sampling: an alternative method for sampling downed coarse woody debris. *Canadian Journal of Forest Research*, **33**, 1564–1579.
- Williams, M. S., Schreuder, H. T., & Terrazas, G. H. 1998. *Poisson sampling—the adjusted and unadjusted estimator*. Research Note RMRS-RN-4. Fort Collins, Colorado: USDA Forest Service.
- Williams, M. S., Valentine, H. T., Gove, J. H., & Ducey, M. J. 2005. Additional results for perpendicular distance sampling. *Canadian Journal of Forest Research*, **35**, 961–966.
- Wolf, A. T., Burk, T. E., & Isebrands, J. G. 1995. Estimation of daily and seasonal whole-tree photosynthesis using Monte Carlo integration techniques. *Canadian Journal of Forest Research*, **25**, 253–260.
- Wolter, K. M. 1985. *Introduction to Variance Estimation*. New York: Springer-Verlag.
- Wood, G. B., & Wiant, H. V. 1992. Inventorying a forest using point-list sampling during a single visit to the field. Pages 397–403 of: Wood, G., & Turner, B. (eds), *Integrating Forest Information Over Space and Time*. Proceedings of IUFROM S4.01 & S4.02 conference. Canberra, Australia: ANUTECH Pty. Ltd.
- Yandle, D. O., & White, F. M. 1977. An application of two-stage forest sampling. *Southern Journal of Applied Forestry*, **1**(3), 27–32.
- Yang, Y.-C., & Chao, S.-L. 1987. Comparison of volume growth calculation methods for remeasured horizontal line sampling. *Forest Science*, **33**, 1062–1067.
- Yates, F. 1949. *Sampling Methods for Censuses and Surveys*. London: Charles Griffin & Co.
- Yu, P. L. H., & Lam, K. 1997. Regression estimator in ranked set sampling. *Biometrics*, **53**, 1070–1080.
- Zeide, B. 1980. Plot size optimization. *Forest Science*, **26**(2), 251–257.

- Zhang, L.-C. 2000. Post-stratification and calibration—a synthesis. *Journal of the American Statistical Association*, **54**, 178–184.

